INCH-POUND

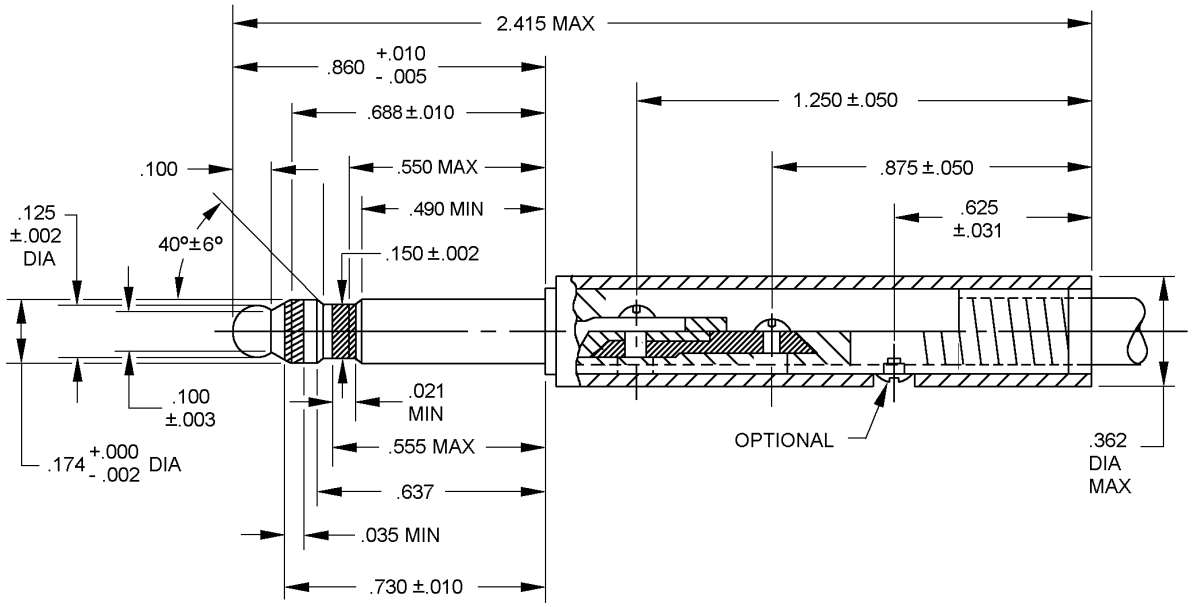
MIL-DTL-642/13C
20 June 2014
SUPERSEDING
MIL-P-642/13B
31 October 1988

DETAIL SPECIFICATION SHEET

# PLUGS, TELEPHONE (TYPES PJ-777 AND PJ-778) 3 CONDUCTOR, SINGLE, DOUBLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-642.

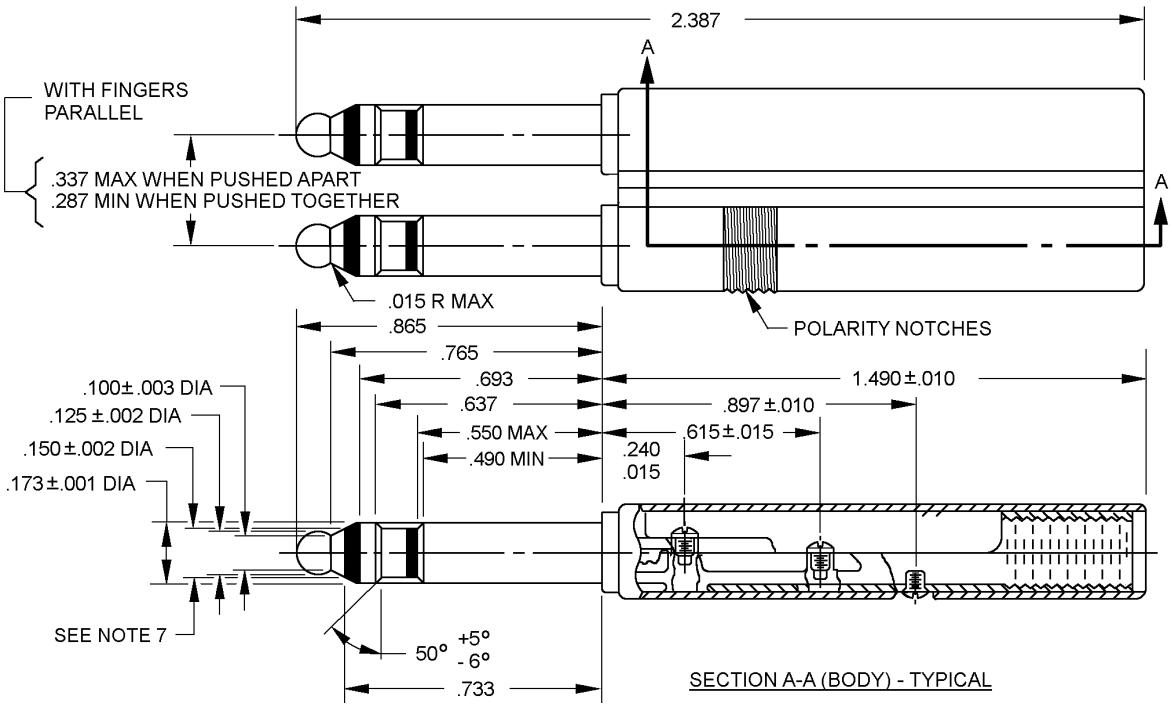



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerance is ±.005 (.127mm) and ±1° on angles.
4. The material shall be brass.
5. All dimensions are after plating.
6. Concentricity shall be .003 inch (.08mm) total indicator reading (TIR).
7. During one cycle of insertion and withdraw test, there will be no shorting or crossover between contacts.

FIGURE 1. Plug PJ-777.



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| Inches | mm | Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|---|---|
| .001 | 0.02 | .125 | 3.18 | .497 | 12.63 | .765 | 19.43 |
| .002 | 0.05 | .150 | 3.81 | .520 | 13.21 | .865 | 21.97 |
| .003 | 0.08 | .173 | 4.39 | .615 | 15.62 | .897 | 22.79 |
| .010 | 0.25 | .240 | 6.01 | .637 | 16.18 | 1.490 | 37.85 |
| .015 | 0.38 | .287 | 7.29 | .693 | 17.60 | 2.387 | 60.63 |
| .100 | .254 | .337 | 8.56 | .735 | 18.67 | | |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerance is ±.005 (.127mm) and ±1º on angles.
4. The material shall be brass.
5. All dimensions are after plating.
6. Concentricity shall be .003 inch (.08mm) total indicator reading (TIR).
7. The left leader line of this dimension points to the intersection of the ramp and the .105 ±.002 diameter cylinder. This intersection is not necessarily the juncture of the dielectric and the outer conductor barrel.

FIGURE 2. Plug PJ-778.

REQUIREMENTS:

Dimensions and configuration: See figure 1 and figure 2.

Shell shall be plastic in accordance with MIL-DTL-642.

Terminal screws shall be 0-80 UNF-2A in accordance with MIL-DTL-642/12.

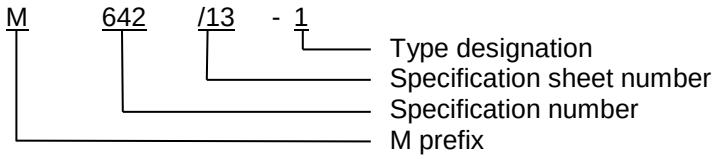
Mating jacks. See table II.



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Supersession data. See table III.

The Part or Identifying Number (PIN) consists of the letter M, the basic specification number, the specification sheet number, and dash number for plug type. (see table I)




PIN example: M642/13-1 indicates type PJ-777B telephone plug.

TABLE I. PIN.

| Dash number | Type designation | Shell material |
|---|---|---|
| 1 | PJ-777B | Black plastic |
| 2 | PJ-777R | Red plastic |
| 3 | PJ-778B | Black plastic |
| 4 | PJ-778R | Red plastic |

TABLE II. Mating jacks.

| Dash number | Mating jacks |
|---|---|
| 1 | M641/19-1 thru -12<br>M641/20-1 thru -8<br>M641/21-1 thru -12 |
| 2 | M641/19-1 thru -12<br>M641/20-1 thru -8<br>M641/21-1 thru -12 |
| 3 | M641/20-1 thru -8<br>M641/21-1 thru -12 |
| 4 | M641/20-1 thru -8<br>M641/21-1 thru -12 |

TABLE III. Supersession data.

| PIN M642/13 | Manufactur's PIN | Manufactur's cage |
|---|---|---|
| -1 | TT - 253 | 82389 |
| -2 | TT – 254 | 82389 |
| -3 | TT – 263 | 82389 |
| -4 | TT - 264 | 82389 |



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Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Referenced documents. In addition to MIL-DTL-642 this document references the following:

MIL-DTL-642/12

## CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – EC
Air Force – 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-2013-055)

Review activities:
Army – AT, AV, CR4, MI
Navy – AS, MC, OS, SH
Air Force – 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil .



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